

Magnetothermodynamics of gadolinium gallium garnet. I. Heat capacity, entropy, magnetic moment from 0.5 to 4.2 °K, with fields to 90 kG along the [100] axis

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Magnetothermodynamics of gadolinium gallium garnet. I. Heat capacity, entropy, magnetic moment from 0.5 to 4.2 °K, with fields to 90 kG along the [100] axis*

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The magnetic moment of a 2.44 cm diam spherical single crystal of gadolinium gallium garnet has been measured with stabilized fields of 500, 1000, 2500, 5000, 10 000, 15 000, 25 000, 40 000, 65 000, and 90 000 G along the [100] axis, over the range 0.35 to 4.2 °K. The heat capacity was measured over similar ranges, except that 500 G was omitted and zero field included. The magnetic moment reached saturation at 0.35 °K and 90 000 G. The electronic and nuclear spin components of the heat capacity at 90 000 G could be separated from each other near 1 °K, thus locating a zero of entropy for the electronic system. By means of 34 adiabatics, corrected to isentropes, which crossed most of the isoerstedic heat capacity series, the entropies as a function of field and temperature were evaluated in terms of the zero reference. The sample has (0.4 ± 0.1) mole% excess entropy over the maximum $R \ln 8$ for Gd^{3+} ion. Thus the sample must have contained 0.4 mole% excess Gd over the stoichiometric amount for $Gd_3Ga_5O_{12}$. Using the correction factor 0.9960, the saturation value for $Gd_3Ga_5O_{12}$ is found to be $39\,095\text{ G} \cdot \text{cm}^3/\text{mole } Gd^{3+}$. This corresponds to a $g = 2.000 \pm 0.002$. Smoothed correlated values of the heat capacity, entropy, enthalpy, internal energy, magnetic moment, and its differential isoerstedic temperature coefficient, differential isothermal magnetic susceptibility, and the isothermal work of magnetization have been tabulated over the ranges 0–90 kG and 0.5–4.2 °K.

Because of their practical applications in magneto optics and in memory and switching devices in computers, synthetic garnets of good quality and large size have recently become available.^{1,2} Synthetic garnets have the general formula $R_3^+M_5^{3+}O_{12}^{2-}$, where R is a rare earth ion and M is most commonly Fe^{3+} , Al^{3+} , or Ga^{3+} . Thus the garnet structure leads to a large number of possible magnetic compounds (and solid solutions) ranging from the relatively simple to the complex.

Single crystals have been made from high purity components and are within better than 99% stoichiometric. They are without visible defects. We selected gadolinium gallium garnet for our first work on this series of compounds. In common with other synthetic garnets, $Gd_3Ga_5O_{12}$ has a cubic crystal structure. A general description of the garnet structure has been given by Goodenough.³ The exact structure for gadolinium gallium garnet has been determined by Brusset, Gillier-Pandraud, and Bordot.⁴ The Gd^{3+} splitting factor, g , has been measured by several groups of investigators^{5–7} using magnetic resonance. Since it is impracticable to make magnetic resonance measurements on the concentrated (i.e., stoichiometric) compound, the investigators used various diamagnetic garnets which were dilute with respect to gadolinium. The most accurate value was found to be 1.992 ± 0.003 .⁷

THE SAMPLE OF GADOLINIUM GALLIUM GARNET

The single crystal sphere used in this research was prepared by Isomet.⁸ They report the follow-

ing procedure: 212.7 g Gd_2O_3 and 187.3 g Ga_2O_3 (each 99.99% pure) were mixed as powders and sintered in a dilute oxygen atmosphere at 1300 °C for 17 h. The above mixture contained a 2.2 mole% excess of Ga_2O_3 over the ideal 5:3 ratio to compensate for loss due to its greater volatility. The sintered mixture was melted at 1690 °C in an iridium crucible using an induction heater. The melting and growth atmosphere was argon, nitrogen, and oxygen in the ratio 25:25:1, i.e., a 2% oxygen atmosphere. Growth was from a seed oriented along the [111] crystal axis, and was at a rate of 5.08 mm/h, with a boule rotation of 15 rpm. Total growth time was about 15 h. The boule was cooled to room temperature at a rate of 50 °C/h. Examination of the boule under crossed polarizers showed very little strain. Other growth imperfections were sought and not found, using Tyndall beam scattering. Samples for analysis were taken from the top and bottom of the boule, which was then ground to a sphere. This was axially oriented by x-rays and marked before drilling alignment holes, all of which was done by Isomet. The holes were 0.5 cm deep and 1.0 mm, rather than the usual 0.6 mm, in diam at opposite ends of each axis. In the present case a 0.6 mm diamond drill was impracticable in the hard garnet. Since our usual Pt9%W aligning pins are only 0.5 mm in diam, in 0.6 mm diam holes, they were fitted with epoxy sleeves before being set in the holes with grease. The three sets of holes were parallel to the [100] axis, a cube edge; a [110] axis, a face diagonal; and a [111] axis, a cube body diagonal.

TABLE I. Emission spectrographic analysis of $\text{Gd}_3\text{Ga}_5\text{O}_{12}$ for trace impurities (analysis supplied by Iso-met).

Z	Element	wt. PPM	At. PPM ^a
12	Mg	2	28
14	Si	5	70
20	Ca	6	51
24	Cr	2	13
26	Fe	5	30
29	Cu	5	27
39	Y	30	114
63	Eu	15	33
82	Pb	2	3
83	Bi	2	3
TOTAL		74	372
Gd^{3+} ~99.96% atomic			

^aAt. PPM $\equiv$ atoms impurity/ 10^6 atoms Gd.

The finished sphere had a diameter of 2.44 cm and weighed 54.1933 g *in vacuo* after the holes had been drilled. This corresponds to 0.16061 moles of Gd^{3+} , assuming stoichiometric $\text{Gd}_3\text{Ga}_5\text{O}_{12}$. The density as calculated from the structural constants⁴ is 7.14 g/cm³ at room temperature.

ANALYSIS OF GADOLINIUM GALLIUM GARNET

The emission spectroscopic analysis supplied by Isomet⁸ is given in Table I, and the spark source

TABLE II. Spark source mass spectrographic analysis of $\text{Gd}_3\text{Ga}_5\text{O}_{12}$ ^{a,b} (analysis by Lawrence Radiation Laboratory, University of California, Livermore, California.)

Z	Element	At. PPM	Z	Element	At. PPM	Z	Element	At. PPM
12	Mg	<10	37	Rb	<3	61	Pm	<1
13	Al	<10	38	Sr	=2	62	Sm	=1
14	Si	<20	39	Y	=10	63	Eu	=7
15	P	<5	40	Zr	<1	64	Gd	MAJOR
16	S	<5	41	Nb	=1	65	Tb	=3
17	Cl	<5	42	Mo	=40	66	Dy	=1
19	K	<5	43	Tc	<1	67	Ho	=2
20	Ca	<10	44	Ru	<2	68	Er	=8
21	Sc	<5	45	Rh	<2	69	Tm	=3
22	Ti	<5	46	Pd	<2	70	Yb	NR
23	V	<5	47	Ag	<2	71	Lu	=10
24	Cr	<10	48	Cd	<2	72	Hf	=10
25	Mn	<5	49	In	NR	73	Ta	<2
26	Fe	<5	50	Sn	=5	74	W	<2
27	Co	<3	51	Sb	<2	75	Re	<2
28	Ni	<3	52	Te	<2	76	Os	<2
29	Cu	=15	53	I	<2	77	Ir	<2
30	Zn	=2	55	Cs	<2	78	Pt	<2
31	Ga	MAJOR	56	Ba	<2	79	Au	<2
32	Ge	<2	57	La	<2	80	Hg	<2
33	As	<5	58	Ce	=2	81	Tl	<2
34	Se	<3	59	Pr	<1	82	Pb	<2
35	Br	<5	60	Nd	=2	83	Bi	<2

114 < atomic ppm impurities < 299

99.97% < Gd^{3+} < 99.99% atomic^aAt. PPM designates atomic parts per million relative to Gd^{3+} .

^b= designates element quantitatively present and determined to an accuracy of $\pm 25\%$. < indicates element not quantitatively determined due to interference, and is present to less than indicated amount. NR indicates element was not read. Elements for $Z < 12$ were not determined.

TABLE III. Magnetic moment for $\text{Gd}_3\text{Ga}_5\text{O}_{12}$ with a magnetic field along the [100] crystal axis. (Observations).^{a,b,c}

T, °K	M	T, °K	M	T, °K	M	T, °K	M	T, °K	M
500 G		2 500 G		10 000 G		25 000 G		40 000 G	
0.329	1 503	0.6465	5 620	1.3132	18 830	0.8038	37 172	3.1898	34 566
0.381	1 425	0.8298	5 439	1.5191	18 204	0.8549	36 973	3.4973	33 884
0.428	1 372	1.0032	5 285	1.7144	17 613	0.9060	36 789	3.7961	33 239
0.5167	1 295	1.1889	5 119	1.8792	17 126	0.9348	36 700	4.0012	32 799
0.6915	1 197	1.3297	4 995	2.0542	16 613	0.9572	36 600	4.1285	32 527
0.8856	1 127	1.5176	4 825	2.2668	16 020	1.0074	36 412		
1.0807	1 073	1.7126	4 653	2.5647	15 226	1.0566	36 224		
1.2624	1 030	1.8727	4 517	2.9128	14 369	1.1061	36 036		
1.4794	983	2.0490	4 368	3.1968	13 727	1.1308	35 955	65 000 G	
1.5124	976	2.2478	4 209	3.5088	13 078	1.1554	35 866		
1.6694	945	2.5697	3 968	3.8029	12 510	1.2048	35 655	0.341	39 112
1.8744	908	2.8852	3 751	4.0014	12 152	1.2550	35 464	0.5435	39 112
2.0812	872	3.1958	3 556	4.1736	11 854	1.3032	35 274	0.6432	39 113
2.2713	841	3.4936	3 385			1.3528	35 082	0.7996	39 110
2.5859	793	3.8021	3 224	15 000 G		1.3555	35 086	0.9419	39 105
2.8680	753	4.0043	3 126			1.4019	34 890	1.1990	39 079
3.1975	712	4.1626	3 052			1.5185	34 437	1.3951	39 040
3.4826	679			0.396	30 994	1.7177	33 665	1.5193	38 982
3.8198	643	5 000 G		0.476	30 745	1.8778	33 055	1.6804	38 923
4.0271	623			0.5641	30 421	2.0522	32 394	1.8719	38 834
4.1750	609			0.6497	30 040	2.2738	31 570	2.0489	38 735
				0.8053	29 299	2.5749	30 487	2.2513	38 609
				0.9710	28 480	2.8987	29 368	2.5663	38 372
				1.1242	27 720	3.1974	28 380	2.8794	38 102
				1.5190	27 394	3.5070	27 406	3.1904	37 807
0.333	2 791	0.6437	10 540	1.2885	26 922	3.7762	26 593	3.4906	37 502
0.386	2 682	0.8017	10 405	1.3314	26 731	3.9641	26 050	3.8007	37 171
0.432	2 606	0.9661	10 237	1.5195	25 832	4.1674	25 480	3.9996	36 969
0.5217	2 494	1.1633	9 991	1.7158	24 960			4.1497	36 780
0.6970	2 344	1.3086	9 785	1.8806	24 269				
0.8921	2 228	1.5185	9 459	2.0534	23 567	40 000 G			
1.0881	2 131	1.7133	9 149	2.2707	22 740				
1.2709	2 051	1.8767	8 896	2.5266	21 815			90 000 G	
1.4901	1 962	2.0526	8 621	2.8912	20 600	0.370	38 765		
1.5183	1 948	2.2376	8 314	3.2029	19 650	0.5023	38 760	0.368	39 252
1.6808	1 887	2.5858	7 884	3.4990	18 815	0.5444	38 760	0.361	39 253
1.8890	1 811	2.8888	7 439	3.7609	18 127	0.6049	38 745	0.5466	39 255
2.0956	1 739	3.1734	7 101	4.0079	17 519	0.6524	38 736	0.6593	39 255
2.2892	1 677	3.4976	6 724	4.1766	17 124	0.7105	38 708	0.8416	39 254
2.6028	1 582	3.8088	6 400			0.8182	38 654	0.9912	39 252
2.8930	1 502	3.9998	6 217			0.8241	38 658	1.2307	39 246
3.2169	1 420	4.1865	6 048	25 000 G		0.9246	38 580	1.4171	39 239
3.5179	1 351					0.9637	38 554	1.5171	39 210
3.8432	1 284			0.299	38 498	1.0310	38 485	1.6776	39 195
4.0728	1 239			0.388	38 357	1.1329	38 376	1.8693	39 173
4.1979	1 219			0.484	38 142	1.1623	38 350	2.0479	39 143
				0.395	20 377	0.5443	37 983	1.3864	38 052
				0.435	20 354	0.5616	37 954	1.5182	37 842
				0.478	20 357	0.5960	37 833	1.6995	37 539
				0.5734	20 351	0.6458	37 701	1.9455	37 228
0.353	6 140	0.6537	20 297	0.6484	37 673	2.0505	36 896	3.5057	38 621
0.405	6 004	0.7994	20 105	0.7014	37 503	2.2502	36 508	3.7999	38 459
0.450	5 907	0.9691	19 760	0.7524	37 332	2.5628	35 873	4.0047	38 338
0.5685	5 714	1.1628	19 261	0.8037	37 155	2.8886	35 189	4.1406	38 255

^aUnits of M are G · cm³/337.45 g sample.^b³He or ⁴He liquid present during measurements.^cValues of T given to less than four places past the decimal were extrapolated and are of reduced accuracy.

mass spectrographic analysis carried out by the Lawrence Radiation Laboratory⁹ is given in Table II.

The samples taken from the upper and lower sections of the boule have been turned over to the analytical group of this department to be checked for a gradient in the gadolinium concentration, but results are not yet available. The highly accurate quantitative analysis of a major constituent, such as gadolinium in the present case, is always difficult. The complete solution of the garnet has not yet been accomplished. Also, the gallium interferes with usual procedures for gadolinium, requiring the preliminary removal of most of the gallium. It is hoped that the analysis can be reported in a later paper in this series.

APPARATUS AND PROCEDURE

Collected references to the apparatus and measuring techniques utilized in this research have

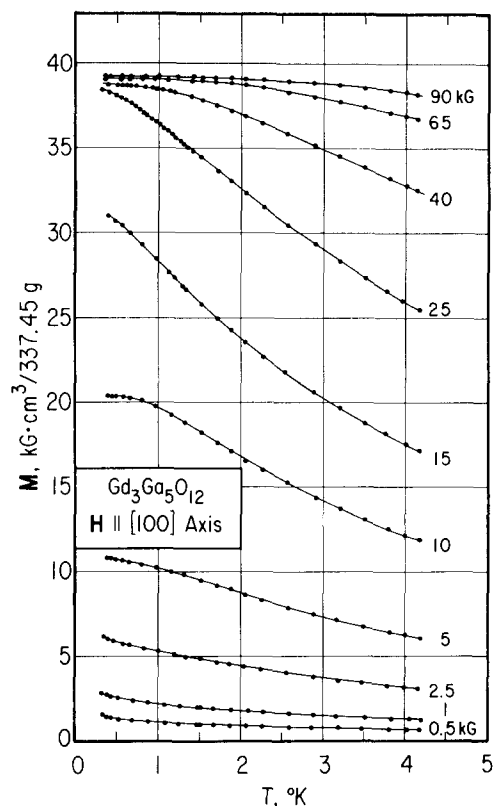


FIG. 1. Magnetic moment of gadolinium gallium garnet with the field along the [100] axis.

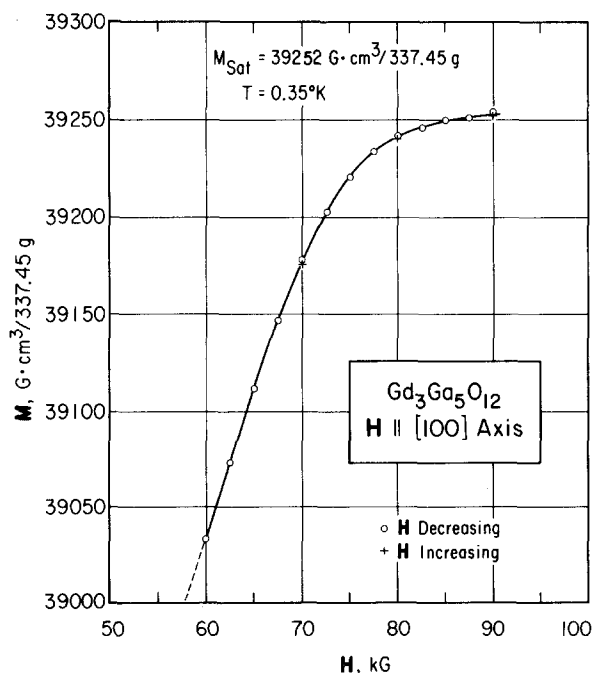


FIG. 2. Magnetic moment of gadolinium gallium garnet sample near saturation at 0.35°K (data uncorrected for lack of stoichiometry).

TABLE IV. Magnetic moments taken in rapid succession at 0.35°K in order to see an approach to saturation moments for $\text{Gd}_3\text{Ga}_5\text{O}_{12}$ with H along the [100] crystal axis. (Observations).^{a,b}

H, G	M^c	$T, ^\circ K$	H, G	M^c	$T, ^\circ K$
90 000	39 254	0.36	70 000	39 178	0.35
87 500	39 251	0.36	67 500	39 147	0.35
85 000	39 250	0.36	65 000	39 112	0.34
82 500	39 246	0.36	62 500	39 073	0.34
80 000	39 242	0.36	60 000	39 033	0.34
77 500	39 234	0.35	70 000	39 177	0.34
75 000	39 221	0.35	80 000	39 241	0.34
72 500	39 203	0.35	90 000	39 252	0.34

^aUnits of M are $G \cdot \text{cm}^3/337.45 \text{ g sample}$.

^b³He gas present in calorimeter during measurements. Pumped ³He liquid in cooling chamber.

^cThese moments were taken some 3 days after those given in Table III, and were 0.05% lower at 65 and 90 kG. This change is within the limit of accuracy of the integrating digital voltmeter used, and accordingly all of the values in this table have been multiplied by 1.0005 to make them consistent with those in Table III.

been given recently¹⁰; comments on our procedures with carbon thermometers have been expanded in a subsequent publication¹¹; and additional magneto-thermodynamic measurements on 7740 Pyrex used in the calorimeter are included in a later paper.¹²

MAGNETIC MOMENT

The magnetic moment was measured by the potentiometric method using an integrating digital voltmeter to measure the $\int E dt$ on surrounding coils. The dimensional geometry of the coils was accurately known with respect to the sample at liq-

TABLE V. Isothermal magnetic moment of $\text{Gd}_3\text{Ga}_5\text{O}_{12}$ with magnetic field along the [100] crystal axis. (Observations).^{a,b}

H, G	M	H, G	M
$T = 1.518^\circ K$			
0	0	12 500	22 233
500	976	15 000	25 832
1 000	1 948	17 500	28 864
1 500	2 916	20 000	31 248
2 000	3 874	25 000	34 437
2 500	4 825	30 000	36 215
3 000	5 768	35 000	37 229
4 000	7 628	40 000	37 842
5 000	9 459	50 000	38 499
6 000	11 260	60 000	38 856
7 000	13 038	65 000	38 981
8 000	14 780	70 000	39 073
9 000	16 512	80 000	39 172
10 000	18 204	90 000	39 209

^aUnits of M are $G \cdot \text{cm}^3/337.45 \text{ g sample}$.

^b⁴He superfluid present during measurements.

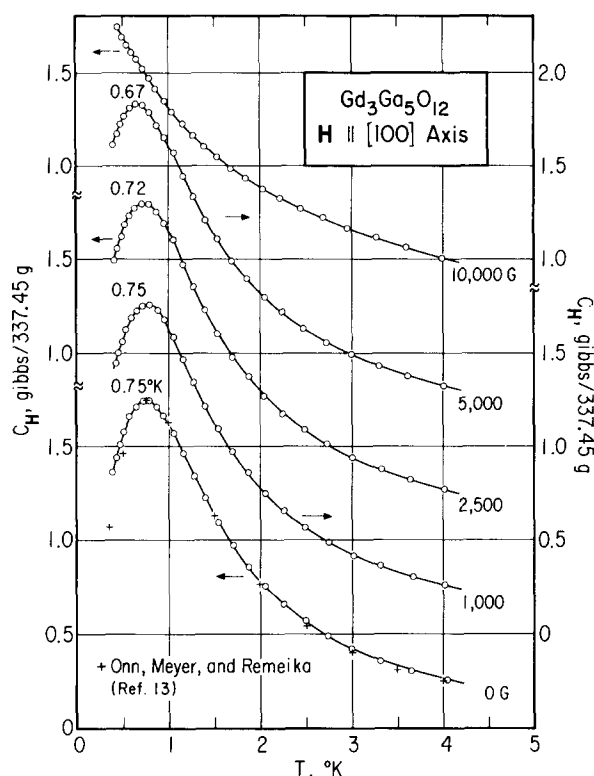


FIG. 3. Heat capacity of gadolinium gallium garnet in the low field region. Field along the [100] axis (axes of ordinates broken).

uid helium temperatures to enable absolute measurements. The coils were split in the middle, and the upper and lower coils could be connected in the same or opposite sense. The most accurate measurements were made by raising or lowering the sample by means of an elevator, with the fixed coils in the opposite sense above and below the coincident equatorial planes of the magnet and the coaxial measuring coils. The magnetic moment measurements were made as the field and temperature were held constant during the calibration of the carbon thermometers, first with liquid ^3He and then with liquid ^4He in contact with the sample. There were carbon thermometers at the top and bottom of the Pyrex calorimeter. Since the bottom thermometer was in close contact with the sample, it was used to record its temperature. The data are given in Table III and shown in Fig. 1.

A set of magnetic moment values taken near saturation at 0.35 °K is given in Table IV and shown in Fig. 2.

An isothermal series of magnetic moments was taken at 1.518 °K at rather close field intervals in order to increase the accuracy of the relative internal energy anchors at the 1.518 °K intersections with the several isoerstedic heat capacity series:

$$\Delta E = T\Delta S + \int_0^H H dM. \quad (1)$$

The determination of the relative entropies of the several isoerstedics, required for use of Eq. (1), will be presented in a later section of this article. The isothermal data are given in Table V.

HEAT CAPACITY

The isoerstedic heat capacity was measured as usual with the Pyrex calorimeter suspended by three thin-wall cross-braced supporting tubes to reduce vibration and produce near thermal isolation. The supporting tubes were hung from a "pod" containing pumped ^3He liquid, held near 0.43 °K. Heat was supplied by a Pt9%W, noninductively wound, wire heater located on the outer wall of the Pyrex calorimeter near the sample. The calorimeter also contained a separate cooling chamber into

TABLE VI. Heat capacity of $\text{Gd}_3\text{Ga}_5\text{O}_{12}$ with a magnetic field along the [100] crystal axis. (Observations).^{a,b}

$T, ^\circ\text{K}$	$C_H(\text{av})$	$T, ^\circ\text{K}$	$C_H(\text{av})$	$T, ^\circ\text{K}$	$C_H(\text{av})$	$T, ^\circ\text{K}$	$C_H(\text{av})$	$T, ^\circ\text{K}$	$C_H(\text{av})$
0 G		2 500 G		10 000 G		25 000 G		65 000 G	
0.338	1.30	0.367	1.45	0.413	2.04	0.360	0.40	0.429	0.009
0.388	1.36	0.398	1.49	0.451	1.74	0.393	0.48	0.491	0.0100
0.436	1.44	0.433	1.55	0.495	1.678	0.435	0.56	0.5442	0.0110
0.476	1.510	0.478	1.618	0.5398	1.6392	0.482	0.652	0.6034	0.0128
0.5224	1.5816	0.5245	1.6783	0.5907	1.6007	0.5332	0.7405	0.6689	0.0165
0.5832	1.6580	0.5742	1.7293	0.6477	1.5585	0.5893	0.8230	0.7411	0.0239
0.6481	1.7132	0.6336	1.7698	0.7126	1.5140	0.6516	0.9024	0.8244	0.0378
0.7139	1.7424	0.7046	1.7877	0.7872	1.4610	0.7197	0.9838	0.9134	0.0604
0.7881	1.7434	0.7839	1.7805	0.8644	1.4049	0.7968	1.0624	1.0088	0.0943
0.8697	1.7127	0.8658	1.7526	0.9474	1.3450	0.8775	1.1327	1.1152	0.1434
0.9542	1.6559	0.9517	1.6917	1.0398	1.2849	0.9643	1.1995	1.2369	0.2115
1.0493	1.5724	1.0455	1.5952	1.1483	1.2210	1.0590	1.2677	1.3677	0.2947
1.1593	1.4613	1.1548	1.4710	1.2654	1.1597	1.1676	1.3101	1.5020	0.3875
1.2782	1.3400	1.2724	1.3456	1.3870	1.1020	1.2835	1.3520	1.6500	0.4918
1.4032	1.2177	1.3958	1.2262	1.5241	1.0433	1.4035	1.3880	1.8127	0.6046
1.5412	1.0934	1.5348	1.1010	1.6764	0.9850	1.5399	1.4187	1.9944	0.7246
1.6951	0.9705	1.6888	0.9820	1.8432	0.9288	1.6892	1.4399	2.1828	0.8384
1.8662	0.8546	1.8583	0.8694	2.0266	0.8755	1.8541	1.4566	2.3889	0.9514
2.0504	0.7504	2.0428	0.7664	2.2271	0.8227	2.0313	1.4649	2.6162	1.0608
2.2555	0.6542	2.2470	0.6723	2.4496	0.7700	2.2274	1.4731	2.8590	1.1596
2.4859	0.5664	2.4769	0.5851	2.6964	0.7163	2.4030	1.4682	3.1234	1.2500
2.7392	0.4862	2.7234	0.5081	2.9646	0.6631	2.6076	1.4634	3.4190	1.3322
3.0147	0.4173	3.0012	0.4370	3.2672	0.6093	2.8903	1.4421	3.7441	1.4073
3.3216	0.3565	3.3187	0.3738	3.5987	0.5558	3.2043	1.4097	4.1107	1.4737
3.6575	0.3038	3.6380	0.3231	3.9776	0.5012	3.5498	1.3652		
4.0542	0.2556	4.0230	0.2749			3.9383	1.3068		
1 000 G		5 000 G		15 000 G		40 000 G		90 000 G	
0.423	1.45	0.405	1.61	0.377	1.19	0.481	0.059	0.6746	0.0032
0.490	1.560	0.442	1.67	0.410	1.14	0.5268	0.0888	0.7507	0.0044
0.5341	1.6230	0.479	1.722	0.446	1.11	0.5885	0.1408	0.8319	0.0059
0.5864	1.6830	0.5201	1.7671	0.487	1.108	0.6566	0.2109	0.9237	0.0088
0.6448	1.7294	0.5763	1.8070	0.5341	1.1138	0.7266	0.2921	1.0240	0.0140
0.7092	1.7540	0.6425	1.8264	0.5866	1.1196	0.8076	0.3917	1.1375	0.0240
0.7852	1.7559	0.7102	1.8178	0.6453	1.1272	0.8911	0.4965	1.2660	0.0422
0.8659	1.7312	0.7861	1.7836	0.7093	1.1351	0.9834	0.6097	1.4012	0.0701
0.9508	1.6761	0.8644	1.7275	0.7849	1.1439	1.0856	0.7285	1.5439	0.1099
1.0450	1.5864	0.9496	1.6516	0.8657	1.1530	1.1923	0.8393	1.7046	0.1659
1.1518	1.4717	1.0445	1.5589	0.9495	1.1594	1.3138	0.9515	1.8789	0.2374
1.2716	1.3464	1.1515	1.4446	1.0424	1.1634	1.4444	1.0570	2.0674	0.3240
1.3994	1.2193	1.2675	1.3280	1.1506	1.1661	1.5842	1.1515	2.2708	0.4231
1.5382	1.0944	1.3902	1.2145	1.2678	1.1565	1.7392	1.2382	2.4950	0.5339
1.6921	0.9721	1.5297	1.0999	1.3894	1.1463	1.9077	1.3165	2.7374	0.6499
1.8654	0.8563	1.6848	0.9889	1.5241	1.1319	2.0878	1.3852	2.9927	0.7646
2.0516	0.7513	1.8529	0.8877	1.6748	1.1145	2.2863	1.4452	3.2617	0.8755
2.2565	0.6559	2.0392	0.7954	1.8409	1.0937	2.5096	1.4982	3.5655	0.9870
2.4843	0.5686	2.2444	0.7070	2.0193	1.0687	2.7521	1.5434	3.9071	1.0917
2.7388	0.4893	2.4688	0.6277	2.2184	1.0384	3.0141	1.5799	4.2718	1.1930
3.0148	0.4201	2.7216	0.5549	2.4397	1.0040	3.3056	1.6056		
3.3214	0.3594	2.9995	0.4864	2.6805	0.9652	3.6276	1.6230		
3.6632	0.3058	3.3037	0.4263	2.9484	0.9215	3.9954	1.6299		
4.0224	0.2616	3.6230	0.3751	3.2511	0.8722				
		4.0034	0.3241	3.5814	0.8211				
				3.9611	0.7604				

^aHeat capacity C_H in gibbs/337.45 g sample. (gibbs = defined calorie/defined deg.)

^bValues of C_H and T given to less than four places past the decimal indicate decreased accuracy due to experimental temperature scale calibration difficulties.

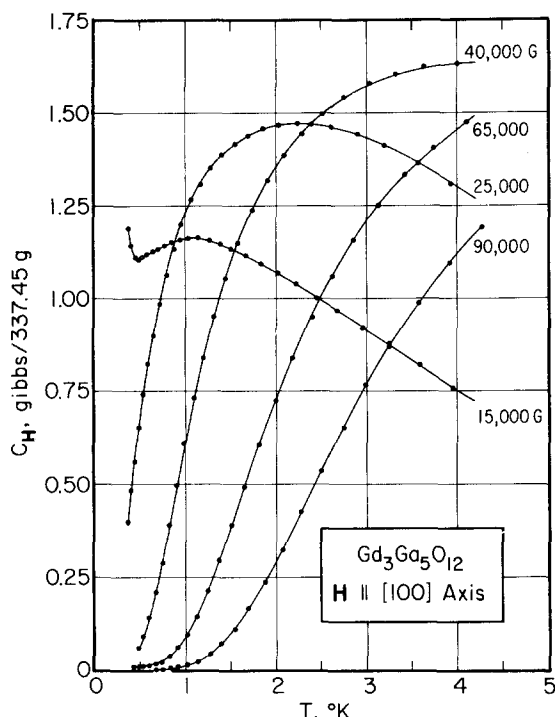


FIG. 4. Heat capacity of gadolinium gallium garnet in the high field region. Field along the [100] axis.

which ^3He liquid could be introduced for preliminary cooling. The conducting gas used in the calorimeter and line was 7.6×10^{-6} moles of ^3He , equal to 1.22×10^{-5} moles of $^3\text{He}/\text{mole Gd}^{3+}$. The unoccupied calorimeter volume was 10 ml and the line volume was 30 ml.

The heat leak between the pod and the calorimeter, plus the vibration input, were determined by a least squares computer analysis of the equilibrium temperature drifts during heat capacity measurements. The best results were obtained in selected regions of low heat capacity. It was represented by

$$\begin{aligned} dQ/dt = & 1.7 \times 10^{-8} + 1.67 \times 10^{-7} (T_{\text{pod}}^2 - T_{\text{cal}}^2) \\ & - 3.88 \times 10^{-8} (T_{\text{pod}}^3 - T_{\text{cal}}^3) \\ & + 8.20 \times 10^{-9} (T_{\text{pod}}^4 - T_{\text{cal}}^4) \text{ cal/sec. } \quad (2) \end{aligned}$$

The constant term in Eq. (2) is due to "vibration." The upper thermometer was used to record the heat leak along the supports. The heat capacity was measured in stabilized fields of 0, 1000, 2500, 5000, 10 000, 15 000, 25 000, 40 000, 65 000, and 90 000 G. The results are collected in Table VI and shown in Fig. 3 (axes of ordinates broken) and in Fig. 4. The zero field curve in Fig. 3 also shows some points taken from the smoothed data in Table IV of Onn, Meyer, and Remeika.¹³ The two sets of

zero-field data are in good agreement. There are no other heat capacity data in constant fields.

These data enable evaluation of the entropy along each isoerstedic with later correlation by means of intersections with temperature-field curves on isentropes.

ADIABATIC CHANGE OF TEMPERATURE WITH MAGNETIC FIELD

Thirty-four series of "adiabatics" which crossed the various constant field heat capacity series were measured using the lower carbon thermometer near the sample. The observations are shown in Fig. 5 and collected in Table VII along with the accumulative entropy corrections needed to correct them to isentropes. These corrections were based on heat leak as given by Eq. (2), plus the minor I^2R heating due to the thermometer currents, divided by the temperature at which the heat was received. A correction due to an exchange of entropy with the Pyrex calorimeter was the major correction. In addition, a correction was made for the variable distribution of the ^3He conduction gas between the calorimeter and line, including the work term when gas entered or left the calorimeter during temperature changes. A derivation¹⁴ and test¹⁵ of the necessary equations has been given previously. Simi-

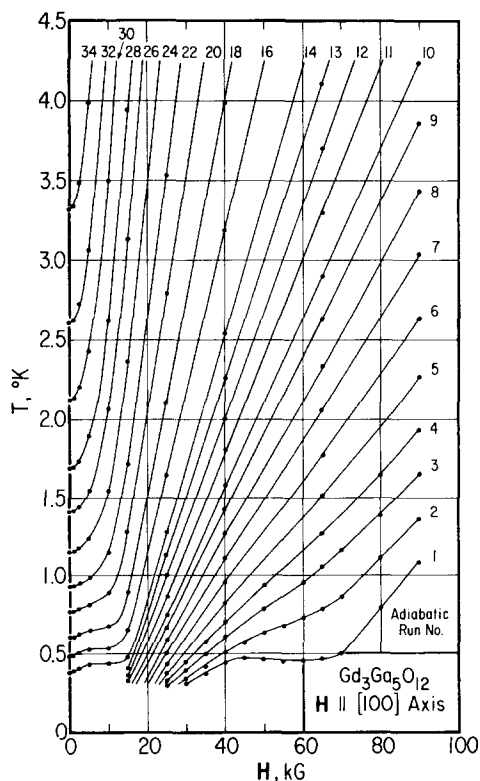


FIG. 5. Temperature vs field on adiabatics for gadolinium gallium garnet with the field along the [100] axis.

TABLE VII. Adiabatic change in temperature with magnetic field along the [100] crystal axis of $\text{Gd}_3\text{Ga}_5\text{O}_{12}$. Small accumulative corrections due to entropy exchange and heat leak are given. (Observations).^{a,b,c}

H,G	T, ^o K	$\Delta S \times 10^3$	H,G	T, ^o K	$\Delta S \times 10^3$	H,G	T, ^o K	$\Delta S \times 10^3$	H,G	T, ^o K	$\Delta S \times 10^3$	H,G	T, ^o K	$\Delta S \times 10^3$	H,G	T, ^o K	$\Delta S \times 10^3$
Series 1			Series 5			Series 12 (contd.)			Series 17 (contd.)			Series 22 (contd.)			Series 28		
90 000	1.0796	0	90 000	2.2584	0	25 000	0.9889	1.2	1 000	0.5466	1.4	1 000	0.9325	1.0	15 000	3.9454	0
70 000	0.488	0.1	65 000	1.5089	0.0	15 000	0.360	1.7	0	0.5432	1.3	0	0.9336	0.9	10 000	2.6233	1.3
65 000	0.463	0.2	40 000	0.8212	0.0	65 000	3.6934	-0.8	40 000	3.5823	-0.6	25 000	3.5308	-0.9	5 000	1.8906	1.4
60 000	0.450	0.3	25 000	0.3693	0.3										2 500	1.7425	1.4
55 000	0.452	0.4	90 000	2.2561	-0.5	Series 13			Series 18			Series 23			1 000	1.6947	1.2
50 000	0.467	0.5				65 000	4.1078	0	40 000	3.9881	0	25 000	3.9395	0	0	1.6938	1.1
45 000	0.469	0.5	Series 6			40 000	2.2633	1.6	25 000	2.1053	1.7	15 000	1.9663	1.7	15 000	3.9393	-1.4
40 000	0.439	0.6	90 000	2.6322	0	25 000	1.1168	1.8	15 000	0.8969	2.2	10 000	1.2981	1.9			
35 000	0.372	0.7	65 000	1.7732	0.1	15 000	0.409	2.2	10 000	0.6696	2.2	5 000	1.0854	1.7	Series 29		
30 000	0.308	1.0	40 000	0.9616	0.1	65 000	4.1055	-1.2	5 000	0.6457	2.1	2 500	1.0399	1.6	10 000	3.0527	0
80 000	0.8412	0.9	25 000	0.431	0.4	Series 14			2 500	0.6227	2.0	1 000	1.0193	1.5	5 000	2.1537	0.4
90 000	1.1237	0.8	90 000	2.6292	-0.8	40 000	2.5422	0	1 000	0.6067	1.9	0	1.0212	1.4	2 500	1.9646	0.3
						25 000	1.2722	0.4	0	0.6044	1.9	25 000	3.9372	-1.1	1 000	1.9061	0.2
Series 2			Series 7			15 000	0.475	0.7	40 000	3.9868	-0.8				0	1.9032	0.1
90 000	1.3609	0	90 000	3.0342	0	10 000	0.433	0.8	Series 19			Series 24			10 000	3.0485	-0.9
80 000	1.1143	-0.0	65 000	2.0546	0.2	5 000	0.431	0.8	25 000	2.4073	0	15 000	2.3566	0	Series 30		
70 000	0.8621	-0.0	40 000	1.1140	0.4	2 500	0.404	0.9	15 000	1.0635	0.6	10 000	1.5459	0.3	10 000	3.5021	0
65 000	0.7842	-0.0	90 000	3.0303	-1.0	1 000	0.384	0.8	10 000	0.7629	0.7	5 000	1.2391	0.2	5 000	2.4324	0.7
60 000	0.7227	-0.0	Series 8			0	0.378	0.9	5 000	0.7195	0.5	2 500	1.1782	0.0	2 500	2.1980	0.7
55 000	0.6735	0.0	90 000	3.4307	0	40 000	2.5426	0.1	2 500	0.6959	0.4	1 000	1.1536	-0.1	1 000	2.1276	0.6
50 000	0.6270	0.0	65 000	2.3313	0.5	Series 15			1 000	0.6799	0.3	0	1.1560	-0.2	0	2.1223	0.4
45 000	0.5709	0.1	40 000	1.2655	0.6	40 000	2.8414	0	0	0.6786	0.3	15 000	2.3554	-0.7	10 000	3.4962	-1.1
40 000	0.5062	0.1	25 000	0.5796	0.8	15 000	1.4410	0.6	25 000	2.4065	-0.3	Series 25			Series 31		
35 000	0.421	0.2	90 000	3.4265	-1.2	10 000	0.5538	0.9	Series 20			15 000	2.7359	0	10 000	3.9798	0
30 000	0.338	0.5	Series 9			5 000	0.476	1.0	25 000	2.7907	0	10 000	1.7949	0.4	5 000	2.7319	1.2
90 000	1.3651	0.1	90 000	3.8630	0	2 500	0.450	1.0	15 000	1.2810	0.8	5 000	1.3900	0.3	2 500	2.4474	1.2
			65 000	2.6322	0.9	1 000	0.432	0.9	10 000	0.8874	0.9	2 500	1.3117	0.2	1 000	2.3631	1.0
80 000	1.3938	-0.0	40 000	1.4323	1.1	0	0.426	0.9	5 000	0.8118	0.7	1 000	1.2825	0.1	0	2.3550	0.8
70 000	1.1517	-0.1	25 000	0.6665	1.2	40 000	2.8418	-0.2	2 500	0.7858	0.6	0	1.2846	-0.0	10 000	3.9703	-1.3
65 000	1.0552	-0.1	90 000	3.8578	-1.5	Series 16			1 000	0.7691	0.5	15 000	2.7338	-0.8	Series 32		
60 000	0.9540	-0.1	Series 10			40 000	3.1916	0	0	0.7686	0.4	Series 26			5 000	3.0653	0
50 000	0.7847	-0.1	90 000	4.2417	0	25 000	1.6414	0.8	25 000	2.7902	-0.5	15 000	3.1371	0	2 500	2.7232	0.1
40 000	0.6005	-0.1	65 000	2.8957	1.4	15 000	0.6527	1.2	Series 21			10 000	2.0652	0.6	1 000	2.6238	-0.0
35 000	0.491	0.0	40 000	1.5794	1.8	10 000	0.5336	1.3	25 000	3.1341	0	5 000	1.5524	0.5	0	2.6127	-0.2
30 000	0.387	0.2	25 000	0.7449	2.0	5 000	0.5286	1.2	15 000	1.4820	0.9	2 500	1.4533	0.4	5 000	3.0593	-0.9
25 000	0.291	0.5	90 000	4.2366	-1.6	2 500	0.5043	1.2	10 000	1.0045	1.1	1 000	1.4185	0.3	Series 33		
90 000	1.6505	0.0	Series 11			1 000	0.4868	1.1	5 000	0.8937	0.9	0	1.4198	0.1	5 000	3.5594	0
			65 000	3.2977	0	0	0.4822	1.1	2 500	0.8634	0.8	15 000	3.1338	-1.0	2 500	3.1322	0.3
80 000	1.9304	0	40 000	1.8048	0.6	40 000	3.1920	-0.4	1 000	0.8459	0.7	Series 27			1 000	3.0078	0.2
70 000	1.6446	-0.1	25 000	0.8666	0.8	Series 17			0	0.8462	0.6	15 000	3.5373	0	0	2.9923	-0.0
65 000	1.2713	-0.1	15 000	0.324	1.4	25 000	3.5827	0	Series 22			10 000	2.3397	0.9	5 000	3.5500	-1.0
50 000	0.9306	-0.1	65 000	3.2967	-0.6	15 000	1.8678	1.1	25 000	3.5325	0	5 000	1.7182	0.9	Series 34		
40 000	0.7003	-0.1	Series 12			10 000	0.7696	1.6	15 000	1.7206	1.3	2 500	1.5959	0.8	5 000	3.9856	0
35 000	0.5707	-0.0	65 000	3.6952	0	5 000	0.5987	1.6	10 000	1.1465	1.4	1 000	1.5550	0.7	2 500	3.4865	0.5
30 000	0.444	0.1	40 000	2.0293	1.0	2 500	0.5863	1.5	5 000	0.9883	1.2	0	1.5551	0.6	1 000	3.3396	0.4
25 000	0.323	0.4							2 500	0.9513	1.1	15 000	3.5332	-1.0	0	3.3198	0.2
90 000	1.9287	-0.3													5 000	3.9720	-1.2

^aUnits of S are gibbs/337.45 g sample (gibbs $\equiv$ defined cal/defined deg).^bValues of T given to less than four places past the decimal were extrapolated and are of sufficient accuracy to enable heat leak corrections.^cSeveral series show a final T greater than the initial T , indicating an entropy gain even though the accumulative correction shows an entropy loss. The discrepancy is less than 0.001 gibbs/mole Gd^{3+} and could indicate some irreversibility.

lar corrections were of course applied to the above heat capacity measurements.

THE MAGNETOTHERMODYNAMIC PROPERTIES OF GADOLINIUM GALLIUM GARNET

In order to locate the zero of electronic entropy it is necessary to resolve the initial rise of the heat capacity curve into its electronic and nuclear components. The most advantageous conditions in the present work are at 90 kG and near 1 °K. The situation is somewhat complicated in the present case because there are two classes of Gd^{3+} ions with respect to a field directed along the [100] axis. They are divided into proportions of 1/3 and 2/3, re-

spectively.¹⁶ The total expression for the fully active heat capacity at 90 kG, including the effects due to nuclear moments, may be expressed as

$$C_H = B_{\text{nuc}} H_{\text{eff}}^2 / T^2 + \frac{1}{3} [(\epsilon_1^2 / RT^2) \exp(-\epsilon_1 / RT)] + \frac{2}{3} [(\epsilon_2^2 / RT^2) \exp(-\epsilon_2 / RT)] + A_L T^3. \quad (3)$$

where B_{nuc} refers to the sum of the nuclear moment effects of the some 30% of the gadolinium represented by ^{155}Gd and ^{157}Gd , together with the two gallium isotopes whose effects have been assumed to be negligible. H_{eff} is the effective field acting on the nuclear moments. The well-known nuclear data are collected in Table VIII. The relatively negligible value of the lattice heat capacity constant, A_L

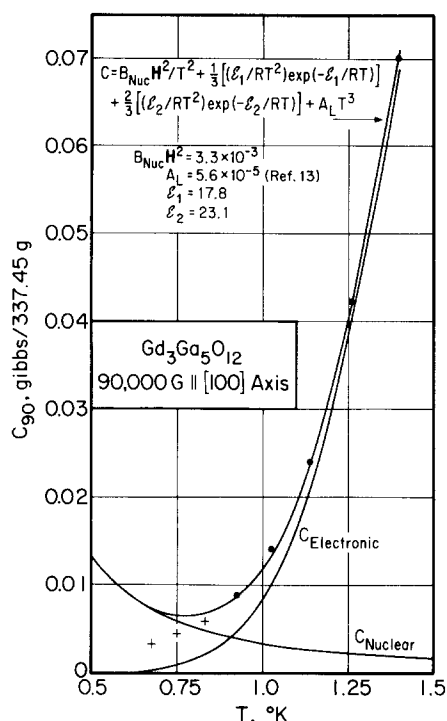


FIG. 6. Heat capacity of gadolinium gallium garnet with a field of 90 000 G along the [100] axis. Data resolved into the electronic, nuclear, and lattice ($\sim$ negligible) components. + Nonequilibrium nuclear moments.

$= 5.6 \times 10^{-5}$ gibbs/deg $^3 \cdot$ mole Gd^{3+} , in the T^3 region was taken from the work of Onn, Meyer, and Remeika.¹³

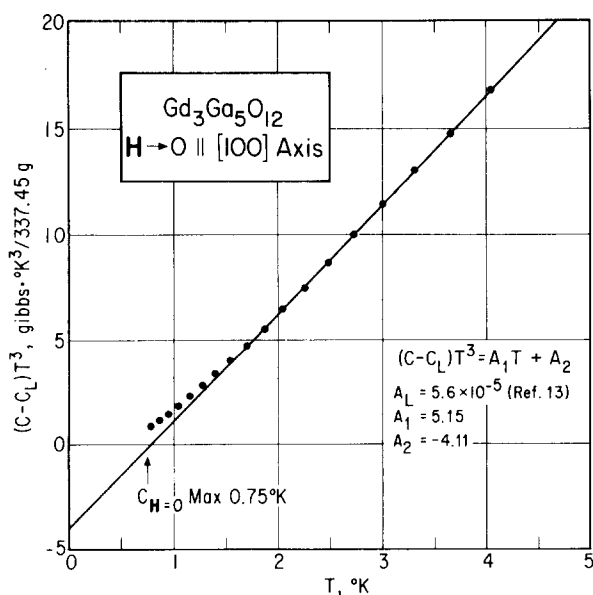


FIG. 7. Plot of $(C_{H=0} - C_{\text{Lattice}})T^3$ vs T used in evaluating the electronic entropy of gadolinium gallium garnet in the region above 4 °K.

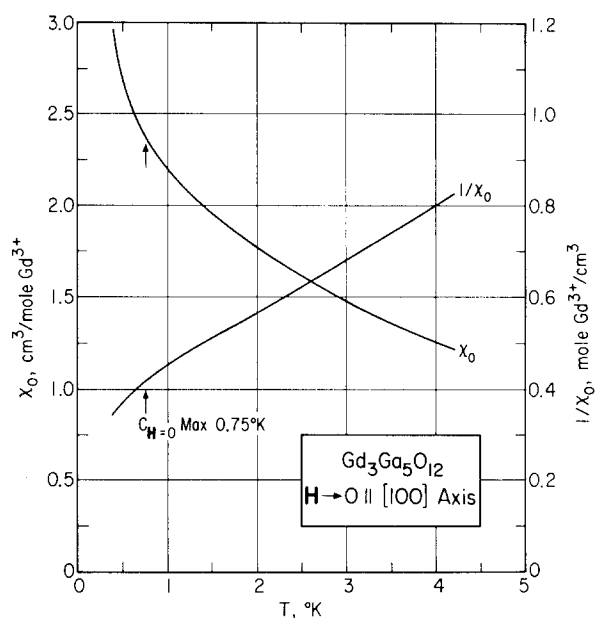


FIG. 8. Initial magnetic susceptibility of gadolinium gallium garnet and its reciprocal vs temperature.

While we believe that Eq. (3) is sufficiently valid to define the limiting exponential decrease of the electronic heat capacity component, the values found for $\epsilon_1 = 17.8$ and $\epsilon_2 = 23.1$ cal/mole Gd^{3+} are too empirical to be taken as very accurate in this concentrated gadolinium compound. The resolved components of the heat capacity at 90 kG, compared with their observed sum, are shown in Fig. 6.

TABLE VIII. Data for computing the fully active ideal nuclear polarization contribution to the heat capacity of $\text{Gd}_3\text{Ga}_5\text{O}_{12}$.^{a,b,c}

Nuclide	% Abund.	I	μ/μ_N	g_I	Moles/Mole Gd^{3+}	B_{Nuc}
⁶⁷ Ga	60.4	3/2	2.0107	1.3405	1.007	6.01×10^{-15}
⁶⁹ Ga	39.6	3/2	2.5549	1.7033	0.660	6.37×10^{-15}
¹⁵⁵ Gd	14.73	3/2	-0.24	-0.16	0.1473	1.25×10^{-17}
¹⁵⁷ Gd	15.68	3/2	-0.32	-0.21	1.1568	2.30×10^{-17}

(Gd 154, 156, 158, 160 all have $I = 0$)

^aGd isotopes 155 and 157 and both isotopes of Ga have nuclear quadrupole moments. The initial nuclear level splittings caused by the quadrupole interactions are assumed small compared to the magnetic field splittings of the levels. The Gd isotopes have a hyperfine interaction taken as $A m_I m_S$.

^b $C_{\text{Nuc}} = B_{\text{Nuc}} H^2/T^2$; $B_{\text{Nuc}} = n[g_I^2 I(I+1)\beta_N^2]/3R$; $W_{\text{Nuc}} = g_I m_I \times \beta_N H_0 + A m_I m_S$ for large H_0 ; $H = H_0 + A m_S/g_I \beta_N$ ($m_S = -7/2$). $R = 8.3151 \times 10^7$ erg/mole $\cdot$ °K and $\beta_N = 3.0412$ erg/mole $\cdot$ G.

^cThe value of A for the hyperfine interaction for ^{155,157}Gd³⁺ is determined by fitting the high field $C_{90\,000}$ G data at low temperature. $B_{\text{Nuc}} H^2 = 3.3 \times 10^{-3}$ which gives $H_{\text{Gd}} = 9.6 \times 10^6$ G $\gg H_0$ with $A_{155} = 0.032$ cal/mole Gd^{3+} and $A_{157} = 0.042$ cal/mole Gd^{3+} . (Alternatively, a transferred hyperfine interaction with the Ga³⁺ nuclei leads to $H_{\text{Ga}} = 5 \times 10^5$ G, which seems unlikely.)

TABLE IX. Magnetothermodynamic properties of gadolinium gallium garnet with magnetic fields along the [100] crystal axis. a, b, c

$T, ^\circ K$	C_H	S	$(H-E_{ref})$	$(E-E_{ref})$	M	$(\partial M/\partial T)_H$	$\int_0^H HdM$	$T, ^\circ K$	C_H	S	$(H-E_{ref})$	$(E-E_{ref})$	M	$(\partial M/\partial T)_H$	$\int_0^H HdM$
0 G								5 000 G							
0.50	1.5427	1.6658	-0.8423	-0.842	0	0	0	0.50	1.7399	1.5148	-1.5919	-0.323	10 622	-906	0.5952
0.55	1.6134	1.8162	-0.7634	-0.763	0	0	0	0.55	1.7852	1.6829	-1.5037	-0.240	10 577	-878	0.5970
0.60	1.6692	1.9591	-0.6813	-0.681	0	0	0	0.60	1.8112	1.8395	-1.4138	-0.155	10 535	-831	0.5982
0.65	1.7087	2.0944	-0.5967	-0.597	0	0	0	0.65	1.8202	1.9849	-1.3229	-0.069	10 493	-838	0.5990
0.70	1.7328	2.2220	-0.5106	-0.511	0	0	0	0.70	1.8143	2.1196	-1.2320	0.017	10 451	-850	0.5992
0.75	1.7411	2.3419	-0.4237	-0.424	0	0	0	0.75	1.7964	2.2443	-1.1417	0.102	10 408	-863	0.5991
0.80	1.7351	2.4542	-0.3368	-0.337	0	0	0	0.80	1.7684	2.3594	-1.0525	0.186	10 365	-903	0.5987
0.85	1.7170	2.5589	-0.2504	-0.250	0	0	0	0.85	1.7325	2.4655	-0.9650	0.268	10 317	-980	0.5978
0.90	1.6890	2.6563	-0.1652	-0.165	0	0	0	0.90	1.6910	2.5634	-0.8794	0.348	10 267	-1 033	0.5964
0.95	1.6533	2.7467	-0.0816	-0.082	0	0	0	0.95	1.6452	2.6536	-0.7959	0.425	10 214	-1 095	0.5947
1.00	1.6116	2.8304	0.0000	0.000	0	0	0	1.00	1.5972	2.7368	-0.7149	0.499	10 157	-1 162	0.5926
1.10	1.5165	2.9797	0.1565	0.157	0	0	0	1.10	1.4940	2.8842	-0.5603	0.639	10 034	-1 283	0.5874
1.20	1.4144	3.1073	0.3031	0.303	0	0	0	1.20	1.3893	3.0097	-0.4161	0.767	9 901	-1 379	0.5810
1.30	1.3126	3.2165	0.4394	0.439	0	0	0	1.30	1.2912	3.1170	-0.2822	0.884	9 759	-1 464	0.5739
1.40	1.2151	3.3101	0.5657	0.566	0	0	0	1.40	1.2004	3.2093	-0.1576	0.991	9 608	-1 538	0.5660
1.50	1.1238	3.3908	0.6826	0.683	0	0	0	1.50	1.1175	3.2893	-0.0418	1.088	9 452	-1 581	0.5574
1.60	1.0395	3.4606	0.7907	0.791	0	0	0	1.60	1.0425	3.3589	0.0661	1.177	9 292	-1 587	0.5485
1.70	0.9625	3.5213	0.8908	0.891	0	0	0	1.70	0.9747	3.4201	0.1669	1.259	9 135	-1 574	0.5398
1.80	0.8927	3.5743	0.9835	0.983	0	0	0	1.80	0.9135	3.4740	0.2613	1.334	8 978	-1 562	0.5312
1.90	0.8298	3.6208	1.0695	1.070	0	0	0	1.90	0.8582	3.5219	0.3498	1.404	8 823	-1 550	0.5225
2.00	0.7729	3.6619	1.1496	1.150	0	0	0	2.00	0.8079	3.5646	0.4331	1.469	8 668	-1 537	0.5139
2.20	0.6748	3.7308	1.2941	1.294	0	0	0	2.20	0.7205	3.6374	0.5857	1.585	8 365	-1 495	0.4968
2.40	0.5934	3.7860	1.4207	1.421	0	0	0	2.40	0.6470	3.6969	0.7222	1.687	8 071	-1 437	0.4799
2.60	0.5249	3.8307	1.5323	1.532	0	0	0	2.60	0.5846	3.7461	0.8452	1.776	7 791	-1 372	0.4638
2.80	0.4672	3.8674	1.6313	1.631	0	0	0	2.80	0.5310	3.7874	0.9566	1.856	7 522	-1 304	0.4481
3.00	0.4186	3.8979	1.7198	1.720	0	0	0	3.00	0.4845	3.8224	1.0581	1.927	7 272	-1 215	0.4338
3.20	0.3771	3.9236	1.7992	1.799	0	0	0	3.20	0.4438	3.8524	1.1508	1.991	7 032	-1 170	0.4198
3.40	0.3415	3.9453	1.8710	1.871	0	0	0	3.40	0.4080	3.8782	1.2359	2.049	6 804	-1 112	0.4063
3.60	0.3104	3.9640	1.9361	1.936	0	0	0	3.60	0.3762	3.9006	1.3143	2.101	6 587	-1 051	0.3935
3.80	0.2835	3.9800	1.9955	1.995	0	0	0	3.80	0.3480	3.9202	1.3866	2.149	6 383	-987	0.3814
4.00	0.2597	3.9939	2.0497	2.050	0	0	0	4.00	0.3227	3.9374	1.4537	2.194	6 192	-928	0.3702
4.20	0.2388	4.0061	2.0995	2.100	0	0	0	4.20	0.3001	3.9525	1.5159	2.234	6 012	-876	0.3597
1 000 G								10 000 G							
0.50	1.5693	1.6530	-0.8744	-0.814	2 507	-1 149	0.0343	0.50	1.6659	1.5052	-3.4234	1.423	20 276	32	2.3452
0.55	1.6377	1.8059	-0.7941	-0.735	2 454	-985	0.0336	0.55	1.6238	1.6619	-3.3412	1.505	20 275	-107	2.3529
0.60	1.6908	1.9508	-0.7109	-0.653	2 408	-858	0.0330	0.60	1.5870	1.8016	-3.2610	1.581	20 258	-592	2.3565
0.65	1.7258	2.0876	-0.6254	-0.569	2 368	-758	0.0324	0.65	1.5518	1.9273	-3.1825	1.650	20 220	-895	2.3555
0.70	1.7461	2.2164	-0.5385	-0.483	2 332	-683	0.0318	0.70	1.5167	2.0410	-3.1058	1.715	20 168	-1 176	2.3519
0.75	1.7519	2.3371	-0.4510	-0.396	2 299	-628	0.0313	0.75	1.4813	2.1444	-3.0308	1.774	20 102	-1 446	2.3455
0.80	1.7472	2.4501	-0.3635	-0.309	2 269	-582	0.0308	0.80	1.4454	2.2389	-2.9577	1.828	20 024	-1 679	2.3371
0.85	1.7319	2.5556	-0.2765	-0.223	2 241	-547	0.0303	0.85	1.4092	2.3254	-2.8863	1.878	19 934	-1 896	2.3270
0.90	1.7064	2.6539	-0.1905	-0.138	2 214	-520	0.0298	0.90	1.3730	2.4049	-2.8167	1.924	19 835	-2 088	2.3153
0.95	1.6707	2.7453	-0.1060	-0.054	2 189	-502	0.0293	0.95	1.3379	2.4782	-2.7490	1.966	19 725	-2 267	2.3022
1.00	1.6273	2.8299	-0.0235	0.028	2 164	-482	0.0288	1.00	1.3047	2.5460	-2.6829	2.004	19 608	-2 409	2.2881
1.10	1.5219	2.9801	0.1340	0.185	2 117	-454	0.0276	1.10	1.2429	2.6674	-2.5556	2.070	19 355	-2 650	2.2575
1.20	1.4149	3.1080	0.2809	0.330	2 073	-434	0.0265	1.20	1.1873	2.7731	-2.4341	2.126	19 081	-2 828	2.2242
1.30	1.3117	3.2171	0.4172	0.466	2 030	-421	0.0254	1.30	1.1375	2.8662	-2.3179	2.173	18 792	-2 945	2.1894
1.40	1.2135	3.3107	0.5434	0.591	1 989	-410	0.0244	1.40	1.0914	2.9487	-2.2065	2.214	18 494	-3 010	2.1538
1.50	1.1223	3.3913	0.6601	0.707	1 948	-398	0.0234	1.50	1.0485	3.0226	-2.0996	2.248	18 191	-3 032	2.1180
1.60	1.0385	3.4610	0.7681	0.814	1 909	-384	0.0224	1.60	1.0087	3.0889	-1.9967	2.279	17 888	-3 028	2.0826
1.70	0.9624	3.5216	0.8680	0.913	1 872	-369	0.0215	1.70	0.9723	3.1490	-1.8977	2.306	17 587	-3 004	2.0477
1.80	0.8933	3.5746	0.9608	1.005	1 835	-359	0.0206	1.80	0.9388	3.2036	-1.8022	2.330	17 288	-2 966	2.0135
1.90	0.8309	3.6212	1.0469	1.090	1 800	-351	0.0197	1.90	0.9079	3.2535	-1.7098	2.352	16 994	-2 913	1.9801
2.00	0.7744	3.6624	1.1272	1.169	1 765	-343	0.0188	2.00	0.8789	3.2993	-1.6205	2.372	16 706	-2 866	1.9478
2.20	0.6768	3.7315	1.2720	1.313	1 698	-324	0.0171	2.20	0.8258	3.3806	-1.4502	2.407	16 139	-2 779	1.8836
2.40	0.5957	3.7868	1.3990	1.438	1 636	-305	0.0154	2.40	0.7781	3.4503	-1.2898	2.437	15 595	-2 660	1.8221
2.60	0.5278	3.8317	1.5111	1.549	1 577	-287	0.0138	2.60	0.7335	3.5108	-1.1387	2.464	15 075	-2 534	1.7635
2.80	0.4701	3.8686	1.6107	1.647	1 521	-271	0.0123	2.80	0.6921	3.5637	-0.9962	2.489	14 581	-2 410	1.7077
3.00	0.4212	3.8994	1.6997	1.735	1 468	-255	0.0108	3.00	0.6537	3.6101	-0.8617	2.511	14 111	-2 288	1.6545
3.20	0.3796	3.9252	1.7797	1.814	1 419	-240	0.0093	3.20	0.6180	3.6511	-0.7346	2.531	13 665	-2 165	1.6041
3.40	0.3438	3.9471	1.8520	1.885	1 372	-226	0.0078	3.40	0.5845	3.6876	-0.6144	2.551	13 244	-2 050	1.5565
3.60	0.3130	3.9648	1.9176	1.949	1 328	-212	0.0064	3.60	0.5531	3.7201	-0.5006	2.569	12 845	-1 946	1.5112
3.80	0.2861	3.9820	1.9774	2.008	1 287	-200	0.0050	3.80	0.5233	3.7492	-0.3930	2.586	12 466	-1 848	1.4680
4.00	0.2625	3.9961	2.0322	2.062	1 248	-190	0.0037	4.00	0.4959	3.7753	-0.2911	2.602	12 105	-1 755	1.4268
4.20	0.2418	4.0084	2.0826	2.112	1 211	-180	0.0024	4.20	0.4704						

TABLE IX. (Continued)

T, °K	C _H	S	(H-E _{ref})	(E-E _{ref})	M	(∂M/∂T) _H	∫ ₀ ^H HdM	T, °K	C _H	S	(H-E _{ref})	(E-E _{ref})	M	(∂M/∂T) _H	∫ ₀ ^H HdM
25 000 G								65 000 G							
0.50	0.6822	0.2818	-15.5822	7.094	37 952	-2 683	8.6288	0.50	0.0100	0.0010	-52.6398	7.882	38 957	-1	9.5566
0.55	0.7625	0.3506	-15.5460	7.048	37 814	-2 833	8.6180	0.55	0.0110	0.0020	-52.6392	7.882	38 957	-2	9.6435
0.60	0.8341	0.4201	-15.5061	7.002	37 670	-2 976	8.6067	0.60	0.0126	0.0030	-52.6387	7.883	38 957	-3	9.7376
0.65	0.8987	0.4895	-15.4628	6.955	37 518	-3 119	8.5947	0.65	0.0149	0.0041	-52.6380	7.883	38 957	-7	9.8385
0.70	0.9572	0.5582	-15.4163	6.906	37 359	-3 265	8.5814	0.70	0.0189	0.0054	-52.6371	7.883	38 956	-12	9.9454
0.75	1.0106	0.6261	-15.3671	6.856	37 192	-3 398	8.5664	0.75	0.0247	0.0069	-52.6361	7.883	38 956	-19	10.0580
0.80	1.0598	0.6929	-15.3153	6.805	37 020	-3 490	8.5503	0.80	0.0326	0.0087	-52.6346	7.883	38 955	-28	10.1759
0.85	1.1053	0.7586	-15.2612	6.754	36 844	-3 538	8.5345	0.85	0.0429	0.0110	-52.6328	7.882	38 953	-40	10.2981
0.90	1.1473	0.8229	-15.2049	6.703	36 666	-3 612	8.5186	0.90	0.0559	0.0138	-52.6303	7.881	38 951	-54	10.4243
0.95	1.1861	0.8860	-15.1465	6.653	36 484	-3 656	8.5024	0.95	0.0715	0.0172	-52.6271	7.879	38 947	-70	10.5539
1.00	1.2229	0.9478	-15.0863	6.603	36 300	-3 719	8.4859	1.00	0.0899	0.0213	-52.6231	7.877	38 944	-88	10.6862
1.10	1.2840	1.0674	-14.9608	6.503	35 923	-3 828	8.4505	1.10	0.1345	0.0319	-52.6120	7.871	38 933	-130	10.9574
1.20	1.3182	1.1807	-14.8305	6.402	35 534	-3 894	8.4105	1.20	0.1884	0.0459	-52.5959	7.864	38 917	-178	11.2342
1.30	1.3523	1.2876	-14.6970	6.302	35 143	-3 889	8.3699	1.30	0.2493	0.0633	-52.5740	7.854	38 897	-229	11.5134
1.40	1.3813	1.3889	-14.5603	6.207	34 756	-3 868	8.3309	1.40	0.3153	0.0842	-52.5458	7.842	38 871	-282	11.7929
1.50	1.4043	1.4850	-14.4209	6.115	34 370	-3 863	8.2914	1.50	0.3843	0.1083	-52.5109	7.829	38 841	-335	12.0705
1.60	1.4220	1.5762	-14.2796	6.026	33 983	-3 853	8.2503	1.60	0.4547	0.1353	-52.4689	7.815	38 804	-387	12.3449
1.70	1.4359	1.6629	-14.1367	5.939	33 599	-3 835	8.2077	1.70	0.5249	0.1650	-52.4199	7.800	38 763	-438	12.6150
1.80	1.4468	1.7453	-13.9925	5.855	33 216	-3 813	8.1635	1.80	0.5938	0.1970	-52.3640	7.784	38 717	-487	12.8798
1.90	1.4549	1.8237	-13.8474	5.773	32 837	-3 782	8.1179	1.90	0.6608	0.2309	-52.3012	7.768	38 666	-535	13.1389
2.00	1.4605	1.8985	-13.7016	5.694	32 460	-3 747	8.0711	2.00	0.7253	0.2664	-52.2319	7.750	38 610	-580	13.3916
2.20	1.4657	2.0380	-13.4089	5.543	31 719	-3 683	7.9736	2.20	0.8456	0.3413	-52.0746	7.714	38 485	-665	13.8768
2.40	1.4640	2.1655	-13.1158	5.401	30 989	-3 610	7.8692	2.40	0.9540	0.4196	-51.8945	7.675	38 345	-742	14.3340
2.60	1.4565	2.2824	-12.8237	5.267	30 277	-3 518	7.7602	2.60	1.0498	0.4998	-51.6939	7.634	38 189	-812	14.7622
2.80	1.4438	2.3899	-12.5336	5.143	29 583	-3 414	7.6485	2.80	1.1333	0.5808	-51.4754	7.590	38 020	-876	15.1617
3.00	1.4266	2.4890	-12.2464	5.029	28 912	-3 315	7.5360	3.00	1.2058	0.6615	-51.2413	7.544	37 839	-932	15.5331
3.20	1.4051	2.5804	-11.9632	4.921	28 258	-3 217	7.4204	3.20	1.2686	0.7413	-50.9937	7.493	37 648	-982	15.8773
3.40	1.3805	2.6648	-11.6846	4.822	27 626	-3 118	7.3048	3.40	1.3233	0.8199	-50.7344	7.441	37 447	-1 026	15.8773
3.60	1.3531	2.7430	-11.4112	4.728	27 011	-3 011	7.1876	3.60	1.3711	0.8970	-50.4649	7.386	37 238	-1 062	16.4906
3.80	1.3234	2.8154	-11.1435	4.639	26 413	-2 919	7.0691	3.80	1.4130	0.9722	-50.1864	7.329	37 022	-1 093	16.7631
4.00	1.2920	2.8824	-10.8819	4.560	25 843	-2 791	6.9559	4.00	1.4498	1.0457	-49.9000	7.272	36 801	-1 117	17.0154
4.20	1.2596	2.9447	-10.6268	4.488	25 296	-2 676	6.8465	4.20	1.4824	1.1172	-49.6067	7.215	36 576	-1 135	17.2491
40 000 G								90 000 G							
0.50	0.0703	0.0069	-29.4876	7.420	38 606	-84	9.0923	0.50	(0.0130)	(-0.0065)	(-75.9674)	(8.135)	39 098	-0	(9.8133)
0.55	0.1058	0.0152	-29.4832	7.420	38 600	-141	9.1735	0.55	(0.0108)	(-0.0054)	(-75.9668)	(8.135)	39 098	-1	(9.9004)
0.60	0.1506	0.0262	-29.4769	7.418	38 592	-214	9.2585	0.60	(0.0091)	(-0.0045)	(-75.9663)	(8.136)	39 098	-2	(9.9951)
0.65	0.2025	0.0403	-29.4681	7.414	38 579	-302	9.3458	0.65	(0.0078)	(-0.0038)	(-75.9659)	(8.136)	39 098	-3	(10.0962)
0.70	0.2593	0.0574	-29.4565	7.409	38 561	-385	9.4350	0.70	(0.0070)	(-0.0033)	(-75.9655)	(8.136)	39 098	-5	(10.2039)
0.75	0.3189	0.0773	-29.4421	7.403	38 540	-480	9.5250	0.75	(0.0065)	(-0.0028)	(-75.9652)	(8.135)	39 097	-8	(10.3176)
0.80	0.3807	0.0998	-29.4246	7.395	38 514	-568	9.6155	0.80	0.0064	-0.0024	-75.9649	8.135	39 097	-11	10.4366
0.85	0.4432	0.1248	-29.4040	7.386	38 483	-660	9.7059	0.85	0.0068	-0.0020	-75.9646	8.133	39 096	-15	10.5605
0.90	0.5056	0.1519	-29.3803	7.376	38 448	-750	9.7957	0.90	0.0078	-0.0016	-75.9642	8.132	39 095	-19	10.6893
0.95	0.5671	0.1809	-29.3535	7.365	38 408	-835	9.8846	0.95	0.0097	-0.0011	-75.9638	8.130	39 094	-24	10.8221
1.00	0.6274	0.2115	-29.3236	7.353	38 364	-927	9.9721	1.00	0.0122	-0.0006	-75.9632	8.128	39 093	-29	10.9589
1.10	0.7410	0.2767	-29.2551	7.325	38 263	-1 093	10.1419	1.10	0.0198	0.0009	-75.9617	8.122	39 090	-40	11.2420
1.20	0.8442	0.3457	-29.1757	7.293	38 146	-1 240	10.3036	1.20	0.0313	0.0031	-75.9591	8.114	39 085	-52	11.5363
1.30	0.9369	0.4170	-29.0866	7.257	38 016	-1 365	10.4572	1.30	0.0476	0.0062	-75.9552	8.106	39 079	-65	11.8397
1.40	1.0194	0.4895	-28.9887	7.220	37 874	-1 476	10.6027	1.40	0.0691	0.0105	-75.9495	8.096	39 072	-78	12.1499
1.50	1.0926	0.5624	-28.8830	7.179	37 721	-1 578	10.7391	1.50	0.0958	0.0161	-75.9412	8.086	39 063	-92	12.4656
1.60	1.1570	0.6350	-28.7705	7.136	37 559	-1 662	10.8666	1.60	0.1272	0.0233	-75.9301	8.076	39 053	-107	12.7848
1.70	1.2137	0.7069	-28.6519	7.093	37 389	-1 729	10.9865	1.70	0.1630	0.0321	-75.9157	8.066	39 042	-123	13.1066
1.80	1.2639	0.7777	-28.5279	7.049	37 213	-1 789	11.0990	1.80	0.2025	0.0425	-75.8974	8.056	39 029	-140	13.4294
1.90	1.3088	0.8473	-28.3993	7.004	37 031	-1 841	11.2039	1.90	0.2453	0.0546	-75.8751	8.046	39 014	-158	13.7523
2.00	1.3486	0.9154	-28.2664	6.958	36 845	-1 886	11.3018	2.00	0.2907	0.0683	-75.8483	8.037	38 997	-177	14.0743
2.20	1.4157	1.0472	-27.9897	6.867	36 460	-1 963	11.4764	2.20	0.3866	0.1005	-75.7806	8.019	38 958	-220	14.7118
2.40	1.4685	1.1728	-27.7010	6.774	36 061	-2 022	11.6249	2.40	0.4852	0.1383	-75.6935	8.002	38 909	-266	15.3354
2.60	1.5108	1.2920	-27.4030	6.681	35 652	-2 067	11.7490	2.60	0.5826	0.1810	-75.5866	7.984	38 851	-315	15.9404
2.80	1.5448	1.4053	-27.0973	6.588	35 235	-2 101	11.8506	2.80	0.6764	0.2276	-75.4607	7.963	38 783	-364	16.5234
3.00	1.5714	1.5128	-26.7855	6.496	34 812	-2 124	11.9314	3.00	0.7652	0.2774	-75.3164	7.941	38 705	-413	17.0825
3.20	1.5916	1.6149	-26.4691	6.405	34 386	-2 130	11.9933	3.20	0.8481	0.3294	-75.1550	7.915	38 618	-459	17.6166
3.40	1.6061	1.7119	-26.1492	6.318	33 961	-2 147	12.0411	3.40	0.9250	0.3832	-74.9756	7.885	38 552	-502	18.1255
3.60	1.6157	1.8040	-25.8270	6.227	33 528	-2 158	12.0666	3.60	0.9951	0.4381	-74.7854	7.853	38 418	-541	18.6099
3.80	1.6207	1.8915	-25.5033	6.140	33 099	-2 145	12.0806	3.80	1.0568	0.4936	-74.5801	7.818	38 306	-576	19.0708
4.00	1.6235	1.9747	-25.1789	6.055	32 671	-2 136	12.0823	4.00	1.1151	0.5493	-74.3629	7.780	38 188	-606	19.5094
4.20	1.6249	2.0539	-24.8540	5.973	32 245	-2 122	12.0723	4.20	1.1702	0.6050	-74.1343	7.743	38 064	-631	19.9277

^aUnits of C_H and S are gibbs/mole Gd³⁺ (gibbs ≡ defined cal/defined deg); units of (H-E_{ref}), (E-E_{ref}), and ∫₀^H HdM are calories/mole Gd³⁺; units of M are G · cm³/mole Gd³⁺; units of (∂M/∂T)_H are G · cm³/°K · mole Gd³⁺. (H-E_{ref}) = (E-E_{ref}) = 0 at 1 °K and 0 G.

^bAll table values have been multiplied by 0.9960 before entry so that the magnetothermodynamic properties will be for 1 mole of Gd³⁺.

^cValues in parentheses have been extrapolated assuming equilibrium between nuclear, electronic, and lattice systems.

The location of the approach of the electronic heat capacity to zero is also substantiated by the fact that the three points, shown centered about 0.75 °K in Fig. 6, showed a slow approach to equilibrium. This indicates that the nuclear heat capacity is co-incidentally going out of equilibrium with the lattice, and that the three points marked + are not true heat capacity values. The typical coincidence of such

events is well-illustrated by the extensive high field data on α-NiSO₄ · 6H₂O,¹⁷⁻²⁰ as well as in other cases.

The B

to the present values $A_{155} = 3.2 \times 10^{-2}$ cal/mole Gd^{3+} and $A_{157} = 4.2 \times 10^{-2}$ cal/mole Gd^{3+} in this concentrated garnet near 1 °K. The zero field heat capacity curve shown in Fig. 3 has a very broad Schottky-like form and Onn, Meyer, and Remeika¹³ suggest that this could be explained by an increased initial splitting of the 8S ground state of Gd^{3+} (expressed as 4.46 °K overall) in the concentrated garnet. It seems more plausible that the spread in the heat capacity is due to cooperative interaction between the Gd^{3+} ions. This could lead to enhanced effective fields due to larger dipole and/or exchange interactions between the cooperative groups. This would not be possible in dilute solutions of Gd^{3+} . It is also possible that H_{eff} may be different for each of the two classes of Gd^{3+} .

After the sum of the electronic and lattice entropies is established at 90 kG, isentropes originating at 90 kG and intersecting the isoerstedic heat capacity curve at 65 kG can be placed on the zero entropy reference. Since there are ten intersections at 65 kG, an average of the closely-agreeing entropy reference values is obtained. Similarly, the intersections at lower fields provide entropy values on all isoerstedics down to zero field. It is convenient to select 1 °K and zero field as the reference point for internal energy, E , and the previously given isothermal Eq. (1),

$$\Delta E = T\Delta S + \int_0^H H dM, \quad (1)$$

then connects the internal energy reference to each isoerstedic. This calculation was made at 1.518 °K, using the values of the magnetic work, $\int_0^H H dM$, obtained at rather close values of H ,

TABLE X. Initial differential magnetic susceptibility for $\text{Gd}_3\text{Ga}_5\text{O}_{12}$ with H along the [100] crystal axis.^{a,b,c}

$T, ^\circ\text{K}$	$(\partial M/\partial H)_T$	$T, ^\circ\text{K}$	$(\partial M/\partial H)_T$	$T, ^\circ\text{K}$	$(\partial M/\partial H)_T$
$H = 0 \text{ G}$					
0.4	2.941	1.7	1.871	3.0	1.467
0.5	2.702	1.8	1.835	3.1	1.441
0.6	2.539	1.9	1.800	3.2	1.417
0.7	2.421	2.0	1.764	3.3	1.393
0.8	2.331	2.1	1.730	3.4	1.370
0.9	2.257	2.2	1.697	3.5	1.349
1.0	2.194	2.3	1.665	3.6	1.327
1.1	2.137	2.4	1.634	3.7	1.306
1.2	2.087	2.5	1.605	3.8	1.285
1.3	2.039	2.6	1.575	3.9	1.266
1.4	1.995	2.7	1.547	4.0	1.246
1.5	1.951	2.8	1.519	4.1	1.227
1.6	1.910	2.9	1.493	4.2	1.208

^aUnits of $(\partial M/\partial H)_T$ are $\text{cm}^3/\text{mole Gd}^{3+}$.

^bSusceptibility was derived from measurements of M at 500 and 1000 G (Table III).

^cValues in table were multiplied by 0.9960 before entry.

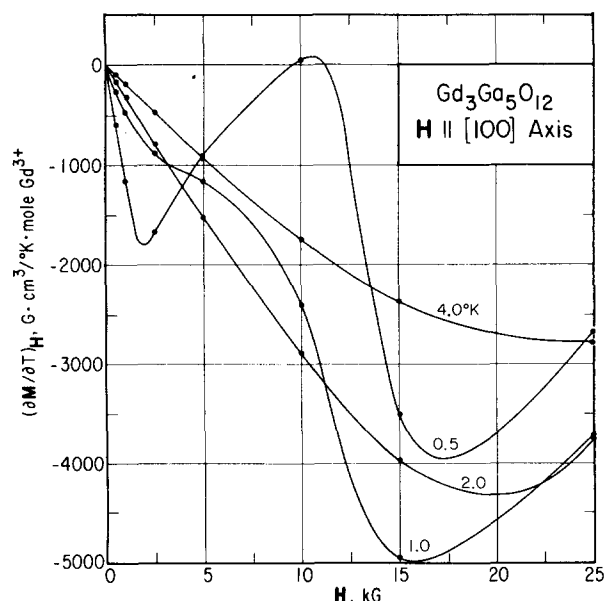


FIG. 9. Temperature coefficient of magnetic moment at constant field vs field along the [100] axis.

which were tabulated above in Table V. The internal energy values were then extended along each isoerstedic by means of the equation

$$\Delta E = \int C_H dT + H\Delta M. \quad (4)$$

The reference point for enthalpy is also 1 °K and zero field, thus

$$(H - H_{\text{ref}})_t = (E - E_{\text{ref}}) - HM. \quad (5)$$

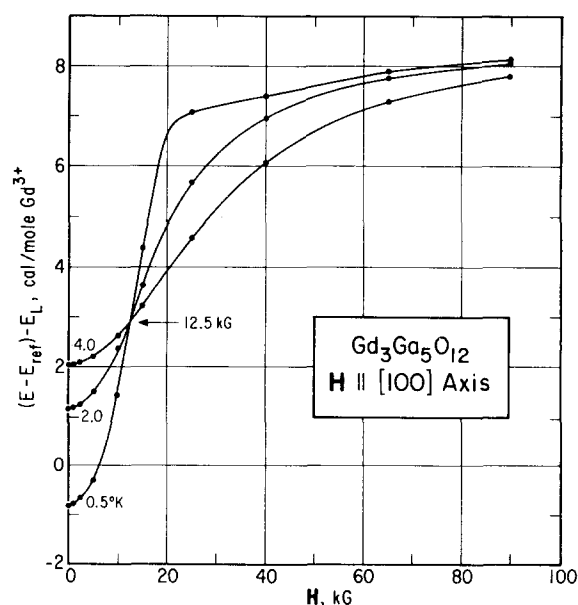


FIG. 10. Relative internal energy on isotherms vs field along the [100] axis.

TABLE XI. Molal magnetic moment and its temperature derivative with constant magnetic field along the [100] crystal axis for $\text{Gd}_3\text{Ga}_5\text{O}_{12}$.^{a,b}

T, °K	M	($\partial M/\partial T$) _H	T, °K	M	($\partial M/\partial T$) _H
H = 500 G					
0.5	1 302	-788	1.6	955	-196
0.55	1 265	-651	1.7	936	-189
0.6	1 237	-538	1.8	917	-183
0.65	1 211	-481	1.9	899	-176
0.7	1 188	-424	2.0	882	-171
0.75	1 168	-383	2.2	849	-162
0.8	1 150	-348	2.4	817	-152
0.85	1 133	-320	2.6	788	-144
0.9	1 118	-299	2.8	760	-135
0.95	1 103	-283	3.0	733	-127
1.0	1 090	-269	3.2	709	-120
1.1	1 064	-247	3.4	685	-113
1.2	1 040	-230	3.6	664	-106
1.3	1 017	-220	3.8	643	-100
1.4	996	-213	4.0	623	-97
1.5	975	-205	4.2	604	-94

^aUnits of M are $\text{G} \cdot \text{cm}^3/\text{mole Gd}^{3+}$; units of ($\partial M/\partial T$)_H are $\text{G} \cdot \text{cm}^3/^\circ\text{K mole Gd}^{3+}$.

^bValues in table were multiplied by 0.9960 before entry.

Also,

$$TdS = dE - HdM, \quad (6)$$

$$= dH + M dH. \quad (7)$$

The correlated data including the temperature coefficient of magnetic moment at constant field, ($\partial M/\partial T$)_H, are given in Table IX.

In estimating the temperature increase in electronic entropy above 4 °K at zero field, we assume that the electronic heat capacity may be expressed by the equation

$$(C_{H=0} - C_L) = (A_1/T^2) + (A_2/T^3). \quad (8)$$

Figure 7 shows a plot of $(C_{H=0} - C_L)T^3$ vs T which evaluates $A_1 = 5.15 \text{ gibbs} \cdot \text{deg}^2/337.45 \text{ g sample}$ and $A_2 = -4.11 \text{ gibbs} \cdot \text{deg}^3/337.45 \text{ g sample}$ by the slope and intercept, respectively. Then, the increase in electronic entropy from $T = 4^\circ\text{K}$ to $T = \infty$ equals

$$\int_4^\infty \frac{(C_{H=0} - C_L)}{T} dT = \frac{A_1}{2T^2} + \frac{A_2}{3T^3} \\ = 0.1395 \text{ gibbs}/337.45 \text{ g sample}. \quad (9)$$

The maximum electronic entropy is,

$$S_{\text{elec}} = S_{4^\circ\text{K}} + \int_4^\infty \frac{(C_{H=0} - C_L)}{T} dT \\ - S_L \text{ (all terms for 337.45 g sample)}, \quad (10)$$

$$= 4.0100 + 0.1395 - 0.0012$$

$$= 4.1483 \text{ gibbs}/337.45 \text{ g sample}. \quad (11)$$

However, the electronic entropy of gadolinium ion cannot exceed $R \ln 8 = 4.1324 \text{ gibbs/mole Gd}^{3+}$. Thus, we conclude that the sample had a small excess of gadolinium due to loss of the more volatile Ga_2O_3 from the melt during the growth of the crystal.

The ratio of gadolinium in 337.45 g sample, over the stoichiometric proportion for $\text{Gd}_3\text{Ga}_5\text{O}_{12}$ is given by $(4.1483/4.1324) = 1.0038$. We will select a provisional factor of 1.004 ± 0.001 and multiply the values of all extensive properties by 0.9960 before entering them in Table IX in order to approximate, as nearly as possible, the values applicable to stoichiometric $\text{Gd}_3\text{Ga}_5\text{O}_{12}$. The data with the field along the [110] and [111] axes, to be considered in later work, should also give factors due to an excess of gadolinium. Hopefully, we will find close agreement, as was the case in β -gadolinium molybdate, $\beta\text{-Gd}_2(\text{MoO}_4)_3$,²² where we also used the electronic entropy as a means of chemical analysis.

The presence of excess gadolinium in the sample also affects the saturation value and the corresponding splitting factor, g . The data given earlier in Table IV and shown in Fig. 2 indicate the approach to saturation at $39\,252 \text{ G} \cdot \text{cm}^3/337.45 \text{ g of sample}$. From this, the saturation value of stoichiometric $\text{Gd}_3\text{Ga}_5\text{O}_{12}$ is estimated as $39\,252 \times 0.9960$ or $M_{\text{sat}[100]} = 39\,095 \text{ G} \cdot \text{cm}^3/\text{mole Gd}^{3+}$. The corresponding value for $g_{[100]} = (M_{\text{sat}}/j\beta) = (39\,095 \times 2/7 \times 5585) = 2.000 \pm 0.002$. This may be compared with the value 1.992 ± 0.003 obtained by Rimai and de Mars⁷ using magnetic resonance on a dilute solution of gadolinium in yttrium gallium garnet at both 300 and 4.2°K .

Values of the initial susceptibility of $\text{Gd}_3\text{Ga}_5\text{O}_{12}$, derived from the data at 500 and 1000 G from Table III, have been given in Table X and shown in Fig. 8. They have been corrected for excess gadolinium in the sample by use of the factor 0.9960.

The values of ($\partial M/\partial T$)_H taken from Table IX are plotted vs field as Fig. 9. The maxima and minima are probably caused by crossing of the states within the eight level manifold of the Gd^{3+} . The relative internal energy on isotherms, also taken from Table IX, are plotted vs field as Fig. 10. A smoothed series of magnetic moments, taken at 500 G, is given in Table XI. The observations were also multiplied by 0.9960.

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